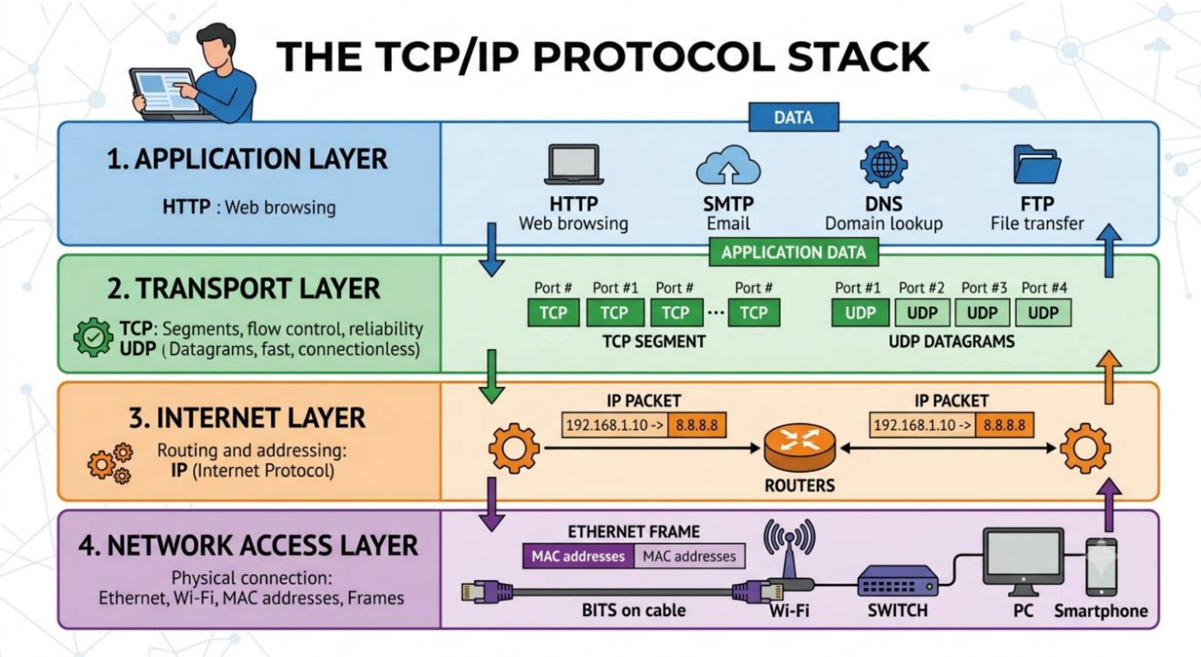
How the Internet Delivers a Web Page

You can use **YouTube** as the example website.

1. TCP/IP Protocol Stack



- Application Layer: Provides services to users. HTTP requests webpages and DNS converts domain names into IP addresses.
- Transport Layer: TCP ensures reliable communication between the browser and server.
- Internet Layer: Uses IP addresses to route packets across networks.
- Network Access Layer: Handles the physical transmission of data through Wi-Fi, Ethernet, etc.

2. Role of DNS

When the user types **www.youtube.com**, the browser does not know the server's IP address.

- Application passes the name to the DNS Resolver.
- Resolver sends a query to the local DNS Server.
- DNS Server returns the corresponding IP address.
- Resolver returns the IP address to the application.

After obtaining the IP address, the browser can communicate with the YouTube server.

3. HTTP Request and Response

HTTP is the protocol used to transfer webpages.

1. Browser creates an HTTP request.
2. The request is sent to the YouTube web server.
3. The server processes the request.
4. The server sends an HTTP response containing the webpage data.
5. The browser displays the webpage to the user.

4. How TCP Provides Reliability

TCP provides reliable communication by:

- Establishing a connection using a three-way handshake.
- Dividing data into segments.
- Numbering each segment.
- Detecting lost packets.
- Retransmitting missing packets.
- Using acknowledgements (ACKs) to confirm delivery.
- Delivering data in the correct order.

Because of TCP, HTTP data reaches the browser accurately and completely.

5.Client–Server or Peer-to-Peer

Here YouTube follows a **client–server architecture**.

Justification:

- The browser acts as the **client**.
- YouTube servers store videos and webpages.
- Clients request services, and servers respond.
- Communication is controlled by centralized servers rather than equal peers.

6. HTTP vs FTP

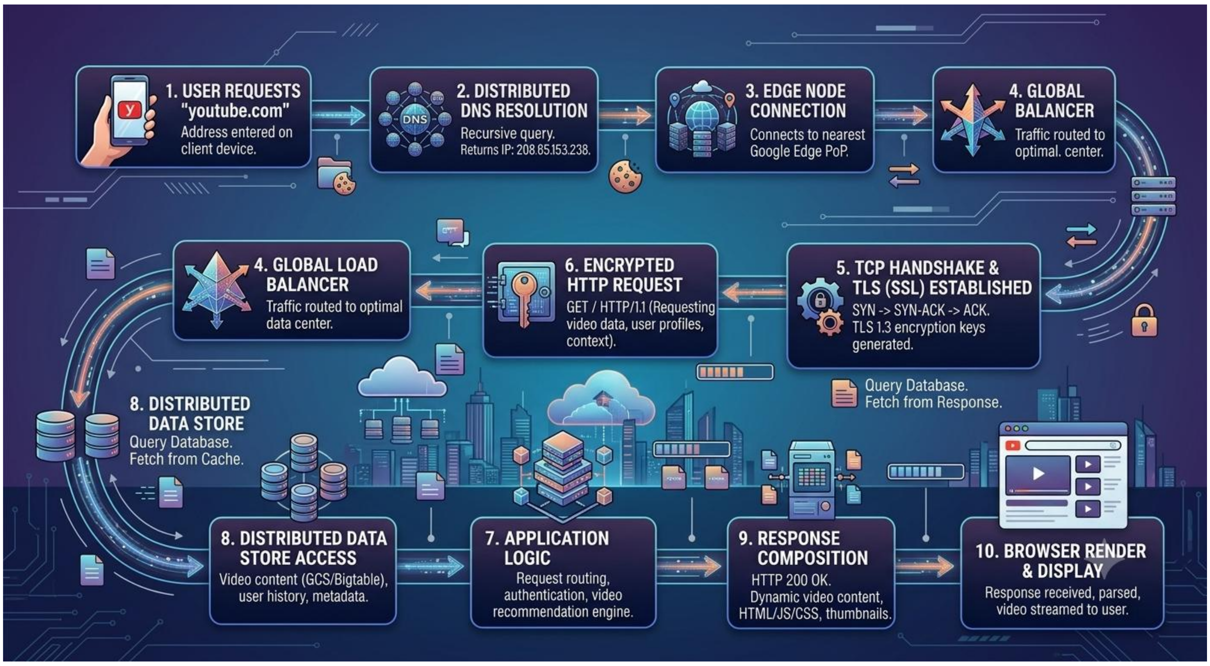
HTTP	**FTP**
Used to transfer web pages	Used to transfer files
Default port is 80	Default ports are 20 and 21
Supports secure communication through **HTTPS (TLS/SSL)**.	Supports secure versions such as **FTPS** and **SFTP**.
Mainly used by web browsers	Mainly by used FTP clients

7. Cookies and User Experience

Cookies store small pieces of information in the user's browser.

Example: When you log in to YouTube and close the browser, cookies remember your login session. The next time you visit YouTube, you remain signed in and receive personalized video recommendations.

8.Flow Diagram



9.Conclusion

Application-layer protocols are essential for everyday Internet communication. Protocols such as HTTP, DNS, FTP, and SMTP allow users to access websites, transfer files, send emails, and use online services efficiently. DNS converts human-readable domain names into IP addresses so that computers can locate servers. HTTP enables browsers to request and receive webpages, while TCP ensures reliable delivery of data. Cookies further improve the user experience by remembering login information and user preferences. Without these protocols, modern Internet services such as YouTube, Google, online banking, social media, and e-commerce would not function properly. Application-layer protocols provide the foundation for communication between clients and servers, making the Internet accessible, reliable, and convenient for billions of users worldwide.